

Assignment

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A CSE260 Assignment



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Problem 1

Assignment Problems: Comprehensive Number Systems Challenge

Problem 1: The Interstellar Probe's Data Log & Onboard Calculations

An autonomous interstellar probe, "Voyager-X," collects vast amounts of data and performs critical calculations onboard. Due to efficiency constraints and legacy systems, the probe uses a mixture of number bases for different data types and arithmetic operations. Your task is to process some of these critical data logs and perform necessary calculations.

- (a) **Navigational Coordinates:** The probe's current celestial coordinates are logged in a unique base-12 system. One specific coordinate is recorded as $(2B.7)_{12}$. Convert this base-12 coordinate to its precise decimal equivalent. Then, take the decimal integer part and convert it to its binary representation, and the decimal fractional part and convert it to its equivalent hexadecimal representation (perform 5-6 steps for the fractional part if it's non-terminating, using dots for the rest).
- (b) **Energy Consumption Metrics (Base Conversion):** The probe's energy consumption is monitored in kilowatt-hours (kWh). A recent spike in activity showed a consumption of $(4785)_{10}$ kWh. For diagnostic purposes, this value needs to be represented in base-7. Convert $(4785)_{10}$ to its base-7 equivalent.
- (c) **Fuel Reserve Management (Arithmetic Operations):** The rover's main fuel tank currently holds $(1A5)_{16}$ units of fuel. A refueling drone delivers an additional $(11011)_2$ units. If the rover then consumes fuel at a rate of $(12)_8$ units per hour for $(10)_5$ hours, and the remaining fuel is subsequently divided equally among $(11)_2$ secondary thrusters, calculate:
- The total fuel after refueling (expressed in base 10).
 - The total fuel consumed during the operation (expressed in base 8).

- The amount of fuel each thruster receives from the remaining fuel (expressed in binary).

Problem 2

Problem 2: Quantum Cryptography Module's Arithmetic & Error Handling

A newly developed quantum cryptography module performs its complex calculations using fixed-bit signed number representations to ensure data integrity and security. Understanding these operations and potential overflows is crucial for debugging.

- Key Generation Algorithm (Subtraction using 2's Complement):** During a key generation cycle, the module needs to calculate the difference between two large prime numbers. Suppose the values involved are 125 and 67. Using an **8-bit two's complement system**, perform the subtraction $125 - 67$. Show all steps, including how you obtain the two's complement of the subtrahend. Justify whether an overflow occurred during this operation.
- Secure Data Encoding (Addition with Negative using 1's Complement):** For secure data encoding, a specific offset, which can be negative, is added to the data block. Consider adding -83 to 55 . Using a **9-bit one's complement system**, perform the addition $(-83) + 55$. Show the one's complement representation of each number, the addition process, and how you handle the end-around carry (if any). Clearly indicate whether an overflow occurred.
- Error Correction Checksum (Addition using 2's Complement):** An error correction checksum is calculated by adding two positive values. Let these values be 150 and 110. Use a **7-bit two's complement system** to perform the addition $150 + 110$. Show all binary steps. Analyze the result and explain if an overflow occurred and why.